

MD Minhazur Rahman

Machine Learning Engineer · Data Scientist · Automation Specialist

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Summary

Data scientist, ML engineer, and automation builder with an MSc in Data Science (University of Greenwich) and 3+ years of industry data/IT experience. I ship production-grade ML end to end — two models live on GCP Cloud Run (Terraform, keyless CI/CD, monitoring), a Kafka stream scorer proven on Kubernetes in CI, and an orchestrated weekly-retraining platform (Dagster, dbt) — plus human-in-the-loop automation tools (document intake, support triage, RPA, a live careers-page fit scorer) with free deterministic cores and optional LLMs. Co-authored, DOI-indexed preprint. Open to ML Engineer, Data Scientist and Automation roles — on-site in Dhaka or Chattogram, remote, or relocation with sponsorship.

Selected ML Projects (all public; two live APIs)

Real-Time Demand Forecasting — Production MLOps Pipeline

2026

LightGBM · FastAPI · Docker · Terraform · Cloud Run

github.com/minhazda/synthetic-retail-mlops-pipeline

- **Cut forecast error 40.8% MAE / 47.3% RMSE vs. a seasonal-naive baseline** by building a typed, tested LightGBM pipeline with a single shared feature module that eliminates training/serving skew.
- **Deployed a live, public inference API on GCP Cloud Run** provisioned entirely as code (Terraform) with keyless GitHub→GCP CI/CD (Workload Identity Federation — no stored secrets), plus production observability (Prometheus, Grafana, Evidently drift reporting).

Real-Time Card-Fraud Detection — Imbalanced Classification

2026

LightGBM · scikit-learn · FastAPI · Docker · Terraform · Cloud Run

github.com/minhazda/fraud-detection-mlops

- **Reached ROC-AUC 0.90 and PR-AUC 0.49 (~8× the base rate) on held-out data** by handling class imbalance (`scale_pos_weight`) and tuning the decision threshold on a separate validation split.
- **Served real-time scoring with built-in explainability** — engineering features from raw transactions at request time and surfacing the risk signals behind each decision in an interactive demo UI.

Event-Driven Fraud Scoring — Kafka Stream Processing

2026

Kafka/Redpanda · Kubernetes · pydantic · Prometheus

github.com/minhazda/streaming-fraud-detection

- **Built a real-time Kafka/Redpanda stream processor** scoring raw transactions with the fraud model: strict schema validation into a dead-letter queue, an alerts topic, and Prometheus scoring-latency histograms — with zero training/serving skew (a unit test pins stream scores to batch predictions at 10^{-9}).
- **Proved the stack on Kubernetes in CI:** the pipeline unit-tests without a broker via a `MessageBus` protocol, then every push runs it end-to-end against a real Redpanda broker and deploys it onto a kind cluster.

Orchestrated Retraining Platform — Champion/Challenger

2026

Dagster · dbt · DuckDB · LightGBM · GitHub Actions

github.com/minhazda/retail-demand-platform

- **Orchestrated the full data-to-model graph as Dagster assets:** dbt builds a tested star schema on DuckDB (data-quality tests run as a pipeline stage), then a LightGBM challenger retrains weekly on 1M+ real retail transactions via a scheduled CI job.
- **Gated deployment on evidence, not retraining:** a champion/challenger rule promotes a new model only on a $\geq 1\%$ holdout-MAE improvement, with champion metrics versioned in git — and a PySpark twin of the feature frame is parity-tested against pandas to 10^{-9} in CI.

Demand Forecasting on Real Retail Data (UCI Online Retail II)

2026

pandas · LightGBM · DuckDB · time-series

github.com/minhazda/real-retail-forecasting

- **Improved pooled MAE 26.3% over a seasonal-naive baseline (beating it on 47/49 product series)** by documenting defensible cleaning of 1.07M real transactions and reproducing the findings in DuckDB SQL.

Privacy-Preserving RAG Agent — LLM Tool-Calling

2026

LangGraph · ChromaDB · on-device embeddings · FastAPI

github.com/minhazda/privacy-preserving-rag-agent

- **Shipped a tool-calling RAG agent behind a fail-closed privacy guard** (on-device embeddings, PII redaction) and built a two-tier evaluation: a deterministic CI gate plus a Claude LLM-as-judge harness with Langfuse tracing.

A/B Experiment Analysis Toolkit

2026

statistics · CUPED · sequential testing · Hypothesis

github.com/minhazda/experiment-analysis-toolkit

- **Built a typed experimentation toolkit** (power analysis, CUPED variance reduction, always-valid sequential testing) whose test suite verifies the *statistical guarantees themselves*: family-wise false-positive rate under continuous peeking and CI coverage over simulation.

Automation & Agentic Tooling (free core; human-in-the-loop; one live)

job-radar — Careers-Page Fit Scanner (live demo)

2026

Streamlit · httpx · BeautifulSoup · Playwright (optional)

huggingface.co/spaces/Minhazda/job-radar

- **Shipped a live, 100%-free (no-API-key) tool** that reads any company’s careers page and scores every role 0–100 for fit with a transparent rule engine and a plain-English rationale — a deterministic HTTP core with an optional local JS-render toggle and multi-company compare.

Invoice Automation — Document Intake & Validation2026

OCR · Pydantic · DuckDB · Streamlitgithub.com/minhazda/invoice-automation

- **Automated PDF-invoice intake at a 73.3% straight-through rate:** OCR → Pydantic + arithmetic validation → DuckDB ledger, routing every malformed document to a human-in-the-loop exception queue so none is booked incorrectly.

AI Triage Agent, RPA & Workflow Automation2026

LangGraph · Playwright · n8ngithub.com/minhazda/ai-triage-agent

- **Built grounded, auditable automations:** a support-triage agent that drafts replies only from policy (never invents) and queues low-confidence cases for a human, plus a resilient Playwright reconciliation bot (retry/backoff, screenshot-on-failure) and three versioned n8n workflows — all tested in CI.

Professional Experience

GPH Ispat LimitedFeb 2022 – Aug 2024

Officer — Information Technology DepartmentChittagong, Bangladesh

- **Automated departmental reporting and operational workflows** by developing internal Python and SQL tools, reducing manual effort across materials-management and reporting processes.
- **Sustained daily operational continuity and data integrity** for enterprise systems used across departments at one of Bangladesh’s largest publicly listed (DSE) steel manufacturers — proactive maintenance, troubleshooting, and materials-management data review feeding company reporting.

Vcube Soft & Tech2021 – Feb 2022

QA Engineer — Computer Vision DataChittagong, Bangladesh

- **Raised training-data quality for production 3D floor-plan reconstruction models** by curating and validating annotated 360° image datasets, enforcing metadata consistency, and building QA pipelines that caught annotation errors before training.

Education

MSc Data Science — Pass with MeritSep 2024 – Jan 2026

University of GreenwichLondon, United Kingdom

- Dissertation (70%): real-time retail demand forecasting on privacy-preserving synthetic data — 37% MAE reduction; published as a DOI-indexed preprint with T. Vuong (Zenodo).
- Modules: Applied Machine Learning 71%, Cloud Computing & HPC 74%, Big Data Analytics, Statistical Methods for Time Series.

BSc Computer Science & EngineeringJan 2017 – Jan 2021

Chittagong Independent UniversityChittagong, Bangladesh

- Final-year project: rare-event detection in surveillance video using 3D-ResNet18 and a custom CNN (PyTorch) on the UCF Crime dataset.

Technical Skills

Languages	Python, SQL, C++, R
ML / DS	scikit-learn, LightGBM, XGBoost, Random Forest, PyTorch; time-series forecasting, imbalanced classification, feature engineering
MLOps / Cloud	Docker, Kubernetes, Terraform (IaC), GCP Cloud Run, GitHub Actions (CI/CD), MLflow, Prometheus, Grafana, Evidently
Streaming / Pipelines	Kafka (Redpanda), PySpark, Dagster, dbt, DuckDB; schema validation, dead-letter queues, champion/challenger promotion
Automation / RPA	Playwright, n8n, OCR document intake, RAG agents (LangGraph); human-in-the-loop exception queues, idempotent workflows, audit logs
LLM / GenAI	LangGraph, RAG, ChromaDB, LLM-as-judge evaluation, Langfuse
Serving / Data	FastAPI, pandas, NumPy, SQL; rolling-origin cross-validation, model evaluation
Testing / Quality	pytest, Hypothesis (property-based), mypy, ruff; typed, tested, reproducible repos

Publications, Awards & Research

- **Publication:** Vuong, T., & Rahman, M. M. (2026). “Synthetic Retail Time-Series for Privacy-Preserving Demand Forecasting.” Preprint, Zenodo. DOI: [10.5281/zenodo.19479285](https://doi.org/10.5281/zenodo.19479285).
- **HPC research:** parallel 2D Jacobi solver in C++/OpenMP on an Azure HPC cluster — 4.81× speedup on 8 threads.
- **Awards & teaching:** Winner, Game of Presentations 2019; Second Runner-Up, City Bank Bankers’ Hunt 2020 (national case competition); ML instruction to 10+ MSc peers and 4 years of STEM/Python tutoring (15+ students).

Reference: Dr. Tuan Vuong — MSc Dissertation Supervisor, University of Greenwich (T.Vuong@greenwich.ac.uk); further references available on request.